Accelerated SoH Forecasting for Second-Life LFP Cells Using a PyBaMM-Trained Neural Operator and Hybrid Model Selection

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Second-life reuse of retired lithium iron phosphate cells requires rapid prediction of future degradation for each incoming cell. Comprehensive characterisation currently requires approximately four weeks per cell. We develop an accelerated forecasting framework that requires only DCIR, an initial-capacity assessment and the first $K=50$ operating cycles, reducing the per-cell testing window to approximately 11 days, or by about 60%. The framework combines exponential extrapolation with a PyBaMM [1]-trained Neural ODE [3] and selects between them using early-cycle deviation from exponential behaviour.

The Neural ODE was trained on 490 PyBaMM SPMe degradation trajectories generated around seven anchor cells from three LFP suppliers (CALB, EVE, REPT) on Prada2013 chemistry [2], using trajectory-grouped train/validation partitions. The context-holdout residual is the exponential-fit RMSE over the last 20% of the $K=50$ context after fitting on the first 80%; a threshold set on synthetic trajectories alone routes each cell to either forecast (Figure 1).

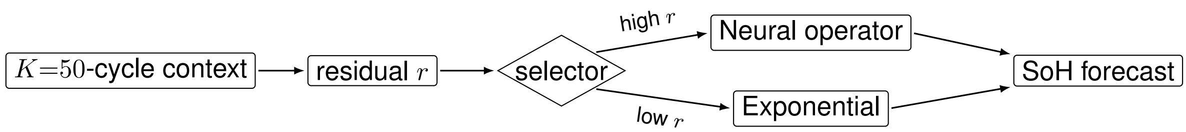

Figure 1: Adaptive selector. A context-holdout residual r (exp fit on the first 80% of the $K=50$ context, RMSE on the last 20%) routes each cell to either the PyBaMM-trained neural operator (high r) or exponential extrapolation (low r); the threshold is set on synthetic trajectories only.

Across three external cells excluded from operator training, exponential and neural forecasts achieved mean SoH RMSEs of 0.70 and 0.80 percentage points, respectively (Figure 2). Across these three cells, the frozen residual-based selector chose the lower-error model for all three and reduced mean RMSE to **0.47 percentage points**, a 33% improvement relative to the best fixed strategy. Exponential extrapolation performed best for CALB_0029 and REPT_0028, and the neural operator performed best for EVE_0003. For REPT_0028, the neural-operator projection extrapolated to ∼2500 cycles remains broadly consistent with the manufacturer's cycle-life specification (≥2500 cycles to 80% capacity), taken as a reference endpoint rather than experimental validation.

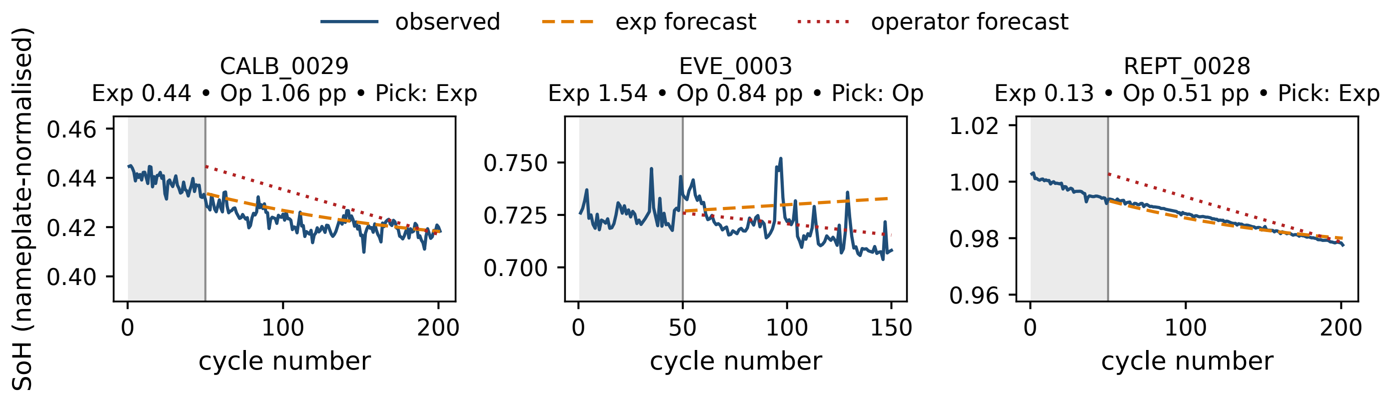

Figure 2: Forecasts evaluated against measured post-context SoH for three external cells. Grey shading denotes the $K=50$ context window and the vertical line marks the transition from context to forecast. Panels use independent y-axis ranges.

Matched ablations showed limited benefit from explicit degradation-parameter conditioning (full-/no-/reduced-θ operators: 0.80, 0.82, 0.90 pp mean RMSE), supporting the framework's premise that matching model choice to early-cycle behaviour is more effective than applying either model uniformly. Reducing the per-cell testing window from ∼28 to ∼11 days could increase laboratory throughput and enable earlier qualification of cells for second-life reuse. Evaluation is limited to three external cells; longer-horizon projections require additional experimental validation.

[1] V. Sulzer et al., *J. Open Res. Softw.* **9**, 14 (2021). • [2] M. Prada et al., *J. Power Sources* **219**, 353 (2012). • [3] R. T. Q. Chen et al., *NeurIPS* (2018).